

Comptia A+ Assignment

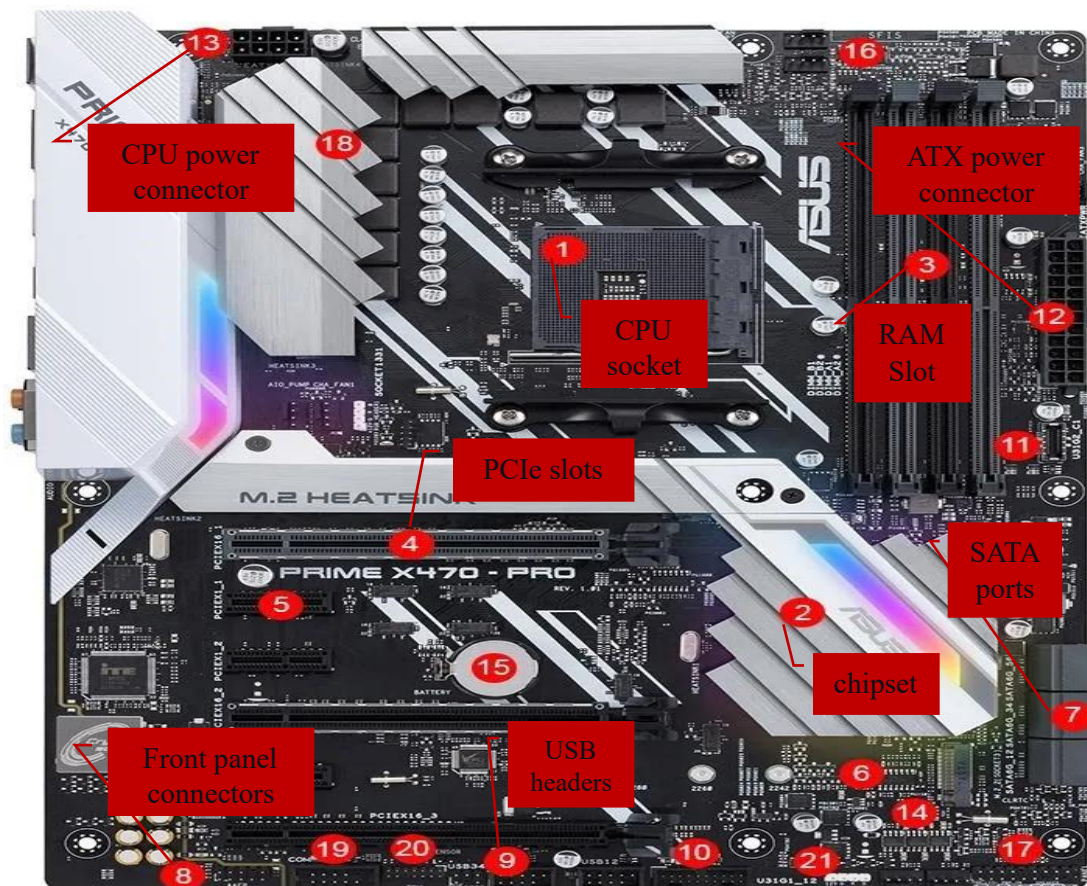
Module -1, 2 [Hardware and its components]

Motherboard and CPU Theory.

Types of motherboards and CPU socket compatibility. Functions of the CPU and chipset. Identify the motherboard slots and ports.

Task

- ❖ 1. Draw a labeled diagram of an ATX motherboard and mark at least 8 different slots and ports (such as RAM slots, PCIe slots, SATA ports, USB headers, CPU socket, power connector, etc.).



❖ **2. Research and list 3 popular CPU socket types (for example: LGA1151, AM4, LGA1200) and name one compatible CPU model for each.**

- A CPU socket is the connector on motherboard where the processor is installed.
- Different CPUs require different socket type, so the socket on the motherboard must match the CPU.
- Below given the CPU socket type and compatibility and their description.

CPU Socket Type	Compatible CPU Model	Description
LGA1151	Intel Core i7-9700K	An Intel socket used for 8th and 9th Generation Intel Core processors. It is commonly found in desktop computers.
AM4	AMD Ryzen 5 5600x	AMD's popular socket that supports many Ryzen processors. It offers excellent performance for gaming and productivity.
LGA1200	Intel Core i5-10400	An Intel socket designed for 10th and 11th Generation Intel Core processors. It provides good performance for everyday computing and gaming.

❖ **3. Explain in your own words the primary functions of the CPU and the chipset on a motherboard, and describe how they work together when you open a music app like Spotify.**

➤ The primary function of the CPU

- The Central Processing Unit (CPU) is known as the brain of the computer.
- It performs calculations, processes instructions, and controls the overall operation of the system.
- Every program and application depends on the CPU to execute commands quickly and accurately.

➤ The primary function of the chipset

- The chipset is an important component on the motherboard that manages communication between the CPU and other hardware devices.
- It controls the flow of data between the processor, memory (RAM), storage devices, USB ports, graphics card, and other peripherals.

➤ How they work together when you open a music app like Spotify

- When you click the Spotify icon, the CPU receives the instruction to open the application. It starts processing the program and asks the chipset to help access the required data.

- The chipset communicates with the SSD or Hard Drive to load Spotify's files into RAM. Once the data is available in RAM, the CPU executes the instructions to launch the application.
- When you play a song, the CPU processes the audio data, while the chipset manages communication between the CPU, RAM, storage, speakers, and internet connection. As a result, Spotify opens smoothly, streams music, and plays sound through the speakers or headphones.

❖ **4. Find a clear photo of a real motherboard (from the internet or your own device), and identify at least 5 different ports or slots by circling them and labeling each one.**

➤ Below given a real motherboard photo with many different ports or slots.

